

N M EMRAN HUSSAIN

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WORK STATUS: [Work permit \(EAD\) Issued](#) | Available Immediately | Relocation available.

SUMMARY

Data Scientist & MSBA graduate specializing in Machine Learning, Generative AI (RAG), Churn Optimization, and Customer Analytics, with a proven track record of translating complex ML pipelines into measurable revenue growth.

EDUCATION

THE GEORGE WASHINGTON UNIVERSITY, School of Business

Washington, DC

Master of Science, Business Analytics

December 2025

Relevant Courses: Responsible Machine Learning, Practical Artificial Intelligence, Working with Large Dataset (Big Data), AI in Marketing, Customer Analytics & Text Analytics (NLP).

NORTH SOUTH UNIVERSITY, School of Business

Dhaka, Bangladesh

Master of Business Administration

May 2023

Relevant Courses: Marketing Research, Brand Management, Global Marketing, Marketing Strategy, Marketing Analytics.

AMERICAN INTERNATIONAL UNIVERSITY - BANGLADESH, School of Engineering

Dhaka, Bangladesh

Bachelor of Science, Electrical and Electronics

March 2009

TECHNICAL SKILLS

Programming & Code Libraries: Python (Pandas, NumPy, Scikit-learn, Matplotlib, TensorFlow, Keras), R (tidyverse, ggplot2)

Machine Learning & AI: Generative AI & LLMs (RAG, Transformers), Deep Learning (TensorFlow/Keras, PyTorch), NLP (Word2Vec, FastText, Sentiment Analysis), Responsible AI (Fairness Metrics, SHAP).

Tools & Platforms: Spark (PySpark), PostgreSQL, ETL Pipelines, Version Control (Git), GitHub, Google Cloud Platform (GCP).

Business Strategy Analytics: A/B Testing, Customer Lifetime Value (CLV), Customer Acquisition Cost (CAC), AARRR Funnel, Churn Prediction, ROMI, ROI Analysis, Stakeholder Management.

ACADEMIC PROJECTS

[Predictive-RAG Retention Engine | Gemini 2.5 Flash, Reinforcement Learning, PySpark ChromaDB, LangChain, GCP](#)

- Engineered AI retention stack (LinUCB/Classification) that cut churn by 34%, driving \$2M in incremental revenue.
- Automated e-commerce retention and decision support via LinUCB Bandits and Gemini-based RAG.

[NLP Sentiment Analysis Optimization | FastText, Python, PySpark, Autotune](#)

- Engineered a sentiment pipeline boosting IMDb accuracy from 50% to 87.4% via optimized supervised learning.
- Standardized 25,000 text samples using FastText Autotune and N-grams to eliminate underfitting and normalize data.

[Loan Default Prediction Model | H2O.ai, Python & Responsible AI](#)

- Compared H2O.ai GLM/ANN models for loan defaults, optimizing a 0.15 threshold to slash financial risk.
- Engineered an L1/L2 pipeline for feature selection and used SHAP to audit model decisions for algorithmic fairness.

[Handwritten Digit Classifier | TensorFlow, Keras & Computer Vision](#)

- Architected a TensorFlow LeNet-5 CNN for MNIST digits, achieving 96.2% accuracy with high validation generalization.
- Optimized a multi-layer pipeline using sigmoid/tanh & MaxPooling to extract spatial features & mitigate vanishing gradients.

[Fairness-Aware Mortgage Pricing Model | Explainable Boosting Machines \(EBM\) & XAI](#)

- Engineered an EBM for high-priced mortgage loans, balancing a 0.78 AUC with federal Fair Lending standards.
- Mitigated bias via AIR analysis and fairness remediation while using extraction attacks to verify security.

[D.C. Bikeshare Optimization & Forecasting | Python & Scikit-Learn](#)

- Forecasted urban mobility demand using ensemble models (GBM, SVM, RF) optimized for cost reduction.
- Optimized bike/dock station balance via K-medoids clustering and multi-model classification.

PROFESSIONAL EXPERIENCE

AMPERE LTD. | Founder & CEO (Business Analyst) | Dhaka, Bangladesh | January 2019 – December 2019

- Leveraged Python and R to model sales trends across Dhaka commercial hubs and simulate freight timelines from China, optimizing inventory mix to maintain a 90% fulfillment rate while reducing overstocking by 5%.
- Built an automated Python pipeline and R dashboard to calculate true unit landed costs (including Bangladesh import taxes) and track supplier quality data, securing a 10% gross margin and accelerating customs clearance.

ANRHUS & SHUZOS CO. | Assistant Manager | Dhaka, Bangladesh | June 2009 – December 2018

- Managed end-to-end supply chain logistics for engineering services, ensuring timely aftersales service delivery.
- Streamlined inventory tracking processes by 5%, reducing overhead costs by 3% & improving operational efficiency by 4%

CERTIFICATION

Google Professional Machine Learning Engineering (on-going)